

Pass-Through

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Grad IO

Pass-Through: What is it?

Lots of cases in economics where we want to know how prices respond to changes in costs:

- ▶ Response to cost shocks (oil shock, commodity prices)
- ▶ Response to tax changes (incidence/efficiency of taxes)
- ▶ Transmission of Exchange Rate Shocks
- ▶ Transmission of Monetary Policy
- ▶ Price effects of Mergers
- ▶ Double Marginalization

Pass-through is the simplest thing in economics that nobody understands

► Often we are talking about different things

- IO: mostly $\rho = \frac{\partial p_j}{\partial mc_j}$ or the matrix $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$
- Macro/Trade: mostly $\frac{\partial \log p_j}{\partial \log mc_j}$
- (Throughout this lecture ρ is the **pass-through rate** – not the nesting parameter from the demand lectures.)

► One thing we know for sure: constant marginal costs, perfect competition:

$$p = mc \rightarrow \frac{\partial p}{\partial mc} = 1$$

► Everything else depends on:

- Curvature of demand
- Nature of competition

Theory of International Trade

- ▶ Most models assume CES and monopolistic competition
- ▶ Implied PT in exchange rate shocks (zero if in local currency, 100% if in foreign currency).

Empirical Results International Trade:

- ▶ Gopinath, Itskhoki, Rigobon (2010): 25% if priced in dollars; 95% if not.
- ▶ Goldberg Hellerstein (2013): around 25% of exchange rate shocks show up in retail beer prices
- ▶ Nakamura Zerom: around 30% of commodity price shocks for coffee show up in retail prices.

- ▶ Tobacco: Harding et al. (2012) $\rho < 1$, while DeCicca et al. (2013) $\rho \approx 1$.
- ▶ Gasoline: Taxes are fully passed through to consumers except when supply is inelastic or inventories were high (Marion and Muehlegger, 2011) but tax holidays $\rho \ll 1$ (Doyle Jr. and Samphantharak, 2008)
- ▶ Sales Taxes: Poterba (1996) found that retail prices of clothing and personal care items $\rho \approx 1$ Besley and Rosen (1999) could not reject $\rho \approx 1$, but found evidence of $\rho > 1$ half of the goods.
- ▶ Alcohol: Young and Bielinska-Kwapisz (2002) $\rho = (1.6, 2.1)$ Kenkel (2005) $\rho = (1.47, 2.1)$

Many studies of “price elasticity” for gasoline, cigarettes, tobacco, etc assume that $\rho = 1$ and regress prices on tax changes to get elasticities: $\frac{\partial \log p}{\partial \log t} = \frac{\partial \log p}{\partial \log c}$ (!)

Recall: Multiproduct Bertrand and Diversion

Recall the Bertrand FOC's for multi-product firm f from the diversion lecture:

$$\rightarrow p_j = q_j(\mathbf{p}) \left[-\frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right]^{-1} + c_j + \underbrace{\sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p}) \left[-\frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right]^{-1}}_{D_{jk}(\mathbf{p})}$$
$$p_j(p_{-j}) = \underbrace{\frac{1}{1 + 1/\epsilon_{jj}(\mathbf{p})}}_{\text{Markup}} \left[c_j + \sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \cdot D_{jk}(\mathbf{p}) \right].$$

Multi-product pricing **raises the opportunity cost** of selling j .

Recall: Upward Pricing Pressure

Agencies often calculate **Upward Pricing Pressure**: UPP asks how the merger **changes** the opportunity cost (recall the UPP frames from the diversion lecture):

$$\left[c_j + \sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \cdot D_{jk}(\mathbf{p}) \right]$$
$$UPP_j = \Delta c_j + \sum_{k \in \mathcal{J}_g} (p_k - c_k) \cdot D_{jk}(\mathbf{p})$$

- ▶ How does the merger change the **opportunity cost** for j ?
- ▶ If the opportunity cost change is positive we say there is **upward pricing pressure**
- ▶ But how much do we expect prices to rise?

- ▶ If we knew the **pass-through rate** $\rho = \frac{\partial p_j}{\partial mc_j}$ we could convert Δmc_j to Δp_j .
- ▶ But we would have to know the **matrix** $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$, which is fine except it is $J \times J$ parameters (again).
- ▶ We can place restrictions on this matrix (maybe diagonal, or maybe just $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}} = 0.8 \cdot \mathbb{I}_J$, etc.).
- ▶ Miller, Ryan, Remer Sheu (2012/2016) did just that: simulate some mergers
 - Calculate UPP
 - What if we had whole $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$ matrix? just the diagonal? just the average diagonal?
- ▶ some approximations are easier than others (depends on curvature of demand)

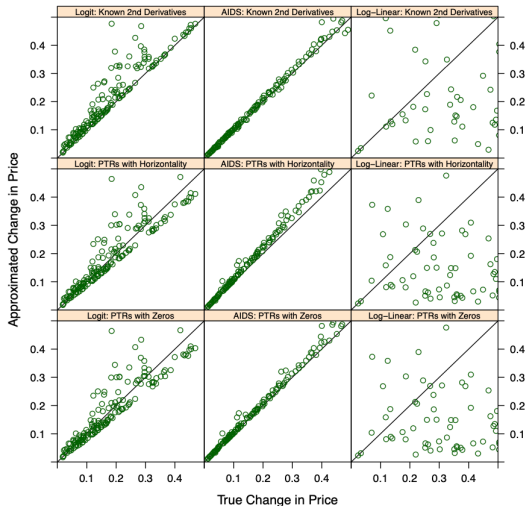


Figure 2: Prediction Error with Complete Information.

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS and log-linear demand. The case of linear demand is omitted because approximation is exact in that setting. Approximations are calculated alternately based on full knowledge of the second-order properties of demand in neighborhood of pre-merger equilibrium ("Known 2nd Derivatives"); based on full knowledge of pre-merger cost pass-through and the horizontality assumption ("PTRs with Horizontality"); and based on full knowledge of pre-merger cost pass-through and the assumption that derivatives of the form $\partial^2 Q_i / \partial P_j \partial P_k$ equal zero ("PTRs with Zeros").

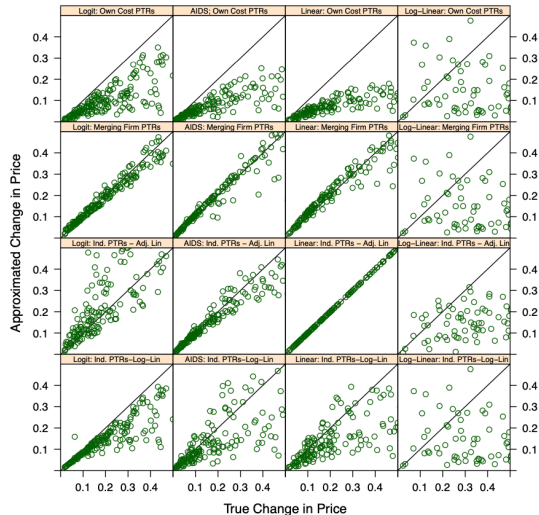


Figure 3: Prediction Error with Incomplete Information

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS, linear demand and log-linear demand. Four informational scenarios are considered: pre-merger cost pass-through that is available only for own costs ("Own Cost PTRs"); pre-merger cost pass-through that is available only for the merging firms ("Merging Firms' PTRs"); industry cost pass-through that is apportioned using the adjusted-linear method ("Ind. PTRs-Adj.-Lin."); and

Perfect Competition: Jenkin (1872), Alfred Marshall (1890)

$$D(p) = S(p - t)$$

$$\rho = \frac{dp}{dt} = \frac{1}{1 + \frac{\varepsilon_d}{\varepsilon_s}}$$

- ▶ Where ε_d is demand elasticity and ε_s is supply elasticity respectively.
- ▶ $\rho \in [0, 1]$ as long demand slopes down and supply slopes up!
- ▶ Constant MC implies that $\varepsilon_s \rightarrow \infty$ and $\rho \rightarrow 1$
- ▶ $I = \frac{\frac{dCS}{dt}}{\frac{dPS}{dt}} = \frac{\rho}{1-\rho}$ (higher pass-through more borne by consumers)

So far so good.

Monopoly (Fabinger Weyl JPE)

Start with $MR = MC$:

$$mr(q) = p(q) + p'(q)q = mc(q) + t$$

$$mr' \frac{dq}{dt} = mc' \frac{dq}{dt} + 1 \Rightarrow \frac{dq}{dt} = \frac{1}{mr' - mc'}$$

$$\Rightarrow \rho = \frac{dp}{dt} = p' \frac{dq}{dt} = \frac{p'}{mr' - mc'}$$

Define monopoly distortion or $ms(q) = p'(q)q$ and

$$\begin{aligned} \rho &= \frac{1}{\frac{p' - ms'}{p'} - \frac{mc'}{p'}} = \frac{1}{1 - \frac{ms' q \cdot p}{q \cdot ms \cdot p'} \frac{q \cdot ms}{q \cdot p} - \frac{mc' q \cdot p}{p' q \cdot mc} \frac{q \cdot mc}{q \cdot p}} \\ &= \frac{1}{1 + \frac{\epsilon_D}{\epsilon_{ms}} \frac{ms}{p} + \frac{\epsilon_D}{\epsilon_S} \frac{mc}{p}}, \end{aligned}$$

Simple right?

Monopoly: Continued (Fabinger Weyl JPE)

Use that $\frac{ms}{p} = -\frac{p'q}{p} = \frac{1}{\epsilon_D}$

and Lerner index $\frac{p-mc}{p} = \frac{1}{\epsilon_D} \Rightarrow \frac{mc}{p} = \frac{\epsilon_D-1}{\epsilon_D}$:

$$\rho = \frac{1}{1 + \frac{\epsilon_D-1}{\epsilon_S} + \frac{1}{\epsilon_{ms}}}$$

- ▶ $\epsilon_D > 1$ for Monopoly (except for my MBA students).
- ▶ $1/\epsilon_{MS} > 0$ log-concave (> 1 concave)
- ▶ $1/\epsilon_{MS} < 0$ log-convex (< 1 convex)

$$\frac{1}{\epsilon_{ms}} = \frac{ms'q}{ms} = \frac{(p''q + p')q}{p'q} = 1 + \frac{p''q}{p'}$$

Can generalize both cases for a conduct parameter $\theta = 0$ (PC); $\theta = 1$ (Monopoly).

$$\rho = \frac{1}{1 + \frac{\theta}{\epsilon_{\theta}} + \frac{\epsilon_D - \theta}{\epsilon_S} + \frac{\theta}{\epsilon_{ms}}}$$

Not sure how I feel about revival of conduct parameter...

A key feature is **log curvature** of demand (second-derivatives)

$$\begin{aligned}(\log D)' &= \frac{D'}{D} = \frac{1}{p'q} = -\frac{1}{ms} \\(\log D)'' &= \frac{ms'}{ms^2} \frac{1}{p'} = -\frac{1}{\epsilon_{ms}} \frac{1}{ms} \left(-\frac{1}{p'q} \right) = -\frac{1}{\epsilon_{ms}} \frac{1}{ms^2}.\end{aligned}$$

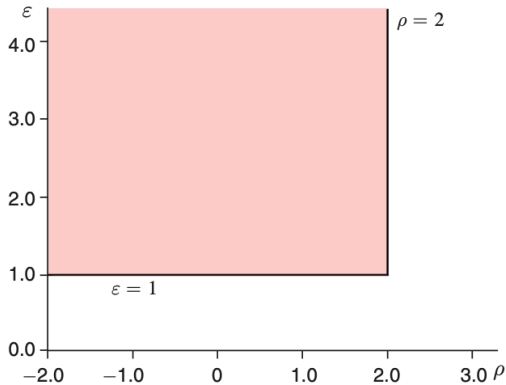
**What if we wrote everything in
terms of inverse demand?**

Just to make things complicated we could express the same ideas using the **inverse demand** $p(q)$ instead of **Marshallian demand** $q(p)$

$$\varepsilon(x) \equiv -\frac{p(x)}{xp'(x)} > 0 \quad \text{and} \quad \rho(x) \equiv -\frac{xp''(x)}{p'(x)}$$
$$\frac{dp}{dc} = \frac{1}{2-\rho} > 0 \Rightarrow \frac{dp}{dc} - 1 = \frac{\rho-1}{2-\rho} \geq 0.$$

- ▶ To be extra-confusing here ρ denotes the curvature of demand, while before it was the pass-through rate.
- ▶ Under monopoly $2p' + xp'' < 0 \Rightarrow \rho < 2$ and $\varepsilon \geq 1$
- ▶ To get $PT > 1$ we need that $\rho > 1$

Panel A. The admissible region



Panel B. The super- and subconvex regions

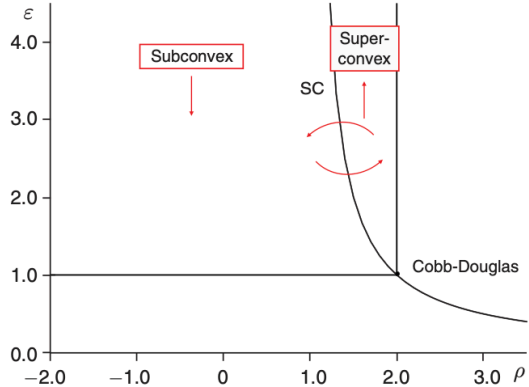
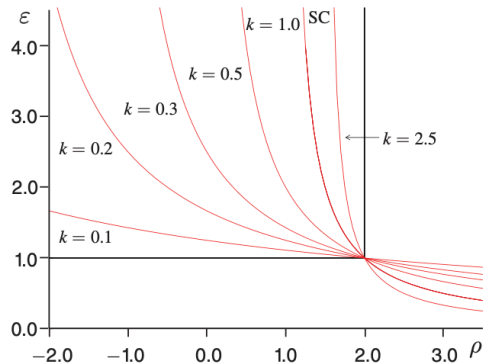


FIGURE 1. THE SPACE OF ELASTICITY AND CONVEXITY

Panel A. Constant proportional pass-through



Panel B. Constant absolute pass-through

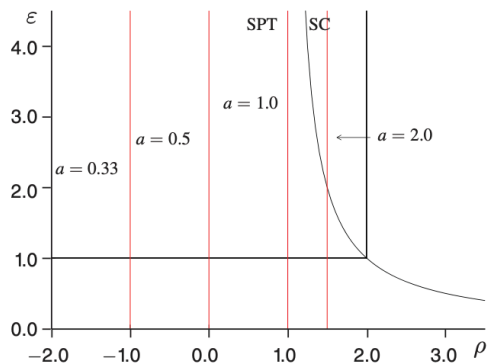


FIGURE 2. LOCI OF CONSTANT PASS-THROUGH

Multiproduct Pass-Through

Multiproduct Pass-Through

- ▶ Most firms sell multiple products.
- ▶ How does that affect what we know about pass-through?
- ▶ Let's start with the case of **double marginalization**.

Retailer and Wholesaler FOC given by:

$$\mathbf{p}^r = \underbrace{\mathbf{p}^w + \mathbf{c}^r}_{\mathbf{mc}^r} + (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$
$$\mathbf{p}^w = \mathbf{mc}^w + \left(\mathcal{H}_w \odot \left(\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} \cdot \Delta_r(\mathbf{p}^r) \right) \right)^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$

- ▶ Δ_r is the matrix of (retail) demand derivatives with entries $-\frac{\partial \sigma_j}{\partial p_k^r}$ (as in the merger simulation lecture).
- ▶ $\mathcal{H}_r, \mathcal{H}_w$ ownership matrix $(j, k) = 1$ if both products sold by same retailer/wholesaler.
- ▶ $\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$ is the **pass-through matrix** (NEW!)

Challenge: We want $\mathbf{p}^r(\mathbf{p}^w)$ and $\mathbf{mc}^w$ but we only have implicit solution for retailer FOC.

How do we get pass-through?

The **pass-through matrix** $\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$ can be obtained in one of two ways:

1. Numerically: perturbing the retailer's marginal costs for each possible choice of k and solving

$$\mathbf{p}^r = \mathbf{mc}^r + e_k + (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$

(Use Morrow Skerlos (2011) formulation and solve for every (j, k) pair).

2. Analytic: Use the retailer's FOC and apply the implicit function theorem.

$$f(\mathbf{p}^r, \mathbf{mc}^r) \equiv \mathbf{p}^r - \mathbf{mc}^r - (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \boldsymbol{\sigma}(\mathbf{p}^r) = 0 \quad (\text{retailer FOC})$$

See Jaffe Weyl (AEJM 2013) or Miller Weinberg (2017 Appendix E) or Conlon Rao (2023).

This is what PyBLP does with `results.compute_passthrough` (very slowly).

The multivariate IFT says that for some system of J nonlinear equations

$$f(\mathbf{p}^r, \mathbf{p}^w) \equiv [F_1(\mathbf{p}^r, \mathbf{p}^w), \dots, F_J(\mathbf{p}^r, \mathbf{p}^w)] = [0, \dots, 0]$$

with J endogenous variables $\mathbf{p}^r$ and J exogenous parameters $\mathbf{p}^w$.

$$\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} = - \begin{pmatrix} \frac{\partial F_1}{\partial p_1^r} & \cdots & \frac{\partial F_1}{\partial p_J^r} \\ \cdots & \cdots & \cdots \\ \frac{\partial F_J}{\partial p_1^r} & \cdots & \frac{\partial F_J}{\partial p_J^r} \end{pmatrix}^{-1} \cdot \underbrace{\begin{pmatrix} \frac{\partial F_1}{\partial p_k^w} \\ \cdots \\ \frac{\partial F_J}{\partial p_k^w} \end{pmatrix}}_{=-\mathbb{I}_J} \quad (\text{PTR})$$

Because the system of equations is additive in $\mathbf{mc}^r = \mathbf{c}^r + \mathbf{p}^w$ this simplifies dramatically.

Use the substitution $\Omega(\mathbf{p}^r) \equiv \mathcal{H}_r \odot \Delta_r(\mathbf{p}^r)$, and differentiate the retailers' system of FOC's with respect to p_l , to get the $J \times J$ matrix with columns l given by:

$$\frac{\partial f(\mathbf{p}^r, \mathbf{p}^w)}{\partial p_l^r} \equiv e_l + \Omega^{-1}(\mathbf{p}^r) \left[\mathcal{H}_r \odot \frac{\partial \Delta_r(\mathbf{p}^r)}{\partial p_l^r} \right] \Omega^{-1}(\mathbf{p}^r) \boldsymbol{\sigma}(\mathbf{p}^r) - \Omega^{-1}(\mathbf{p}^r) \frac{\partial \boldsymbol{\sigma}(\mathbf{p}^r)}{\partial p_l^r}. \quad (1)$$

The complicated piece is the demand Hessian: a $J \times J \times J$ tensor with elements (j, k, l) , $\frac{\partial^2 \sigma_j}{\partial p_k^r \partial p_l^r}$, since

$$\frac{\partial \Delta_r(\mathbf{p}^r)}{\partial p_l^r} = -\frac{\partial^2 \boldsymbol{\sigma}}{\partial \mathbf{p}^r \partial p_l^r}.$$

This also shows a key relationship between **pass through** and **demand curvature** (2nd derivatives).

Pass-through Counterfactuals? (Conlon Rao 2023)

How do we solve for p^w under a counterfactual pass-through matrix?

- Idea: pass-through only augments the matrix $\Delta_r(\mathbf{p}^r)$.
- Example: a constant sales tax rate $\mathcal{P} \equiv \frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} = \text{diag}(1 + \tau_r)$

$$\mathbf{p}^w = \mathbf{m}\mathbf{c}^w + \left(\mathcal{H}_w \odot \left(\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} \cdot \Delta_r(\mathbf{p}^r) \right) \right)^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$

Adapt the Morrow Skerlos ζ fixed point where $\mathcal{P}\Delta(\mathbf{p}_t) = \mathcal{P}\Lambda_t(\mathbf{p}_t) - \mathcal{P}\Gamma_t(\mathbf{p}_t)$

$$\mathbf{p}_t \leftrightarrow \mathbf{c}_t + \boldsymbol{\zeta}_t(\mathbf{p}_t) \quad \text{where}$$

$$\boldsymbol{\zeta}_t(\mathbf{p}_t) = \Lambda_t(\mathbf{p}_t)^{-1} \mathcal{P}^{-1} [\mathcal{H}_t^* \odot \mathcal{P} \Gamma_t(\mathbf{p}_t)] (\mathbf{p}_t - \mathbf{c}_t) - \Lambda_t(\mathbf{p}_t)^{-1} \mathcal{P}^{-1} \boldsymbol{\sigma}_t(\mathbf{p}_t)$$

For diagonal $\mathcal{P}$ (not sure about general case with Hadamard product):

$$\boldsymbol{\zeta}_t(\mathbf{p}_t) = \Lambda_t(\mathbf{p}_t)^{-1} [\mathcal{H}_t^* \odot \Gamma_t(\mathbf{p}_t)] (\mathbf{p}_t - \mathbf{c}_t) - \Lambda_t(\mathbf{p}_t)^{-1} \mathcal{P}^{-1} \boldsymbol{\sigma}_t(\mathbf{p}_t)$$

What's the point?

Why do we care about pass-through in vertical settings?

- ▶ When PT is high, we think that downstream firms have significant market power
- ▶ This also means the scope for **elimination of double marginalization** is large.
 - Higher pass-through should be associated with more efficient vertical mergers? (Not sure this has been tested)

What's the point?

- ▶ Because second-derivatives of plain logit are pinned down by price coefficient and shares, we don't have any flexibility in curvature.
- ▶ Mixed logit is less restricted but how much less? (See Miravete Seim Thurk 2023 at CEPR)
- ▶ Probably these models are under-parametrized....but
- ▶ A huge class of models adds an extra parameter between wholesale and retail prices...

Estimating Pass Through: Conlon Rao (AEJP: 2020)

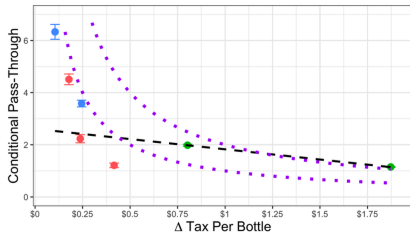
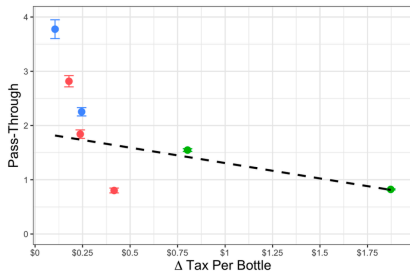
A simple empirical exercise

- ▶ US states tax distilled spirits by volume
- ▶ We see tax changes in three states (Connecticut, Louisiana, Illinois)
- ▶ Three popular bottle sizes (750mL, 1L, 1.75L)
- ▶ Run a regression at a 3 month difference product by product

$$\Delta p_{jst} = \rho_{jst}(\mathbf{X}, \Delta\tau) \cdot \Delta\tau_{jt} + \beta\Delta x_{jst} + \gamma_j + \gamma_t + \epsilon_{jst}$$

- ▶ Only innovation: allow $\rho_{jst}(\mathbf{X}, \Delta\tau)$ to vary by size of tax increase.

Goes horribly wrong

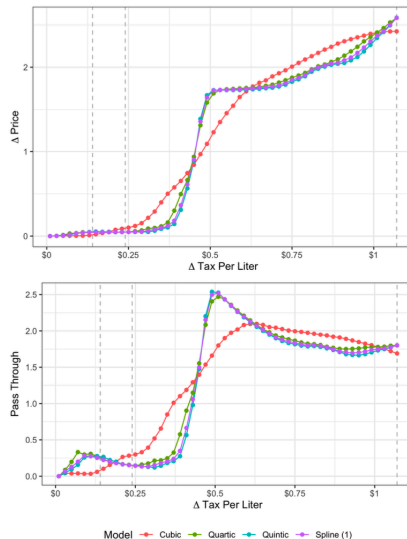


state CT IL LA

- ▶ Regression coefficients are huge > 1 and all over the map
- ▶ Conditional on a price change even larger > 3
- ▶ Seem to coincide with the purple line

Is PT a structural parameter?

- ▶ Price points are a big problem here
- ▶ Estimated an ordered probit with flexible $\beta(\Delta\tau)$
- ▶ Pass-through depends on where you are; not stable at all (!)
- ▶ Maybe structural interpretations of pass-through parameter are a bad idea!



Some challenges

- ▶ Pless and Van Benthem (AEJA: 2019)

Use pass-through > 1 as a test for market power? (Is this true?)

- ▶ Chetty Looney Croft (AER 2016)

$$\log q(p, \tau^S) = \alpha + \beta \log p + \theta_\tau \beta \log(1 + \tau^S)$$

$$\theta_\tau = \frac{\partial \log q}{\partial \log(1 + \tau^S)} / \frac{\partial \log q}{\partial \log p} = \frac{\varepsilon_{q, 1 + \tau^S}}{\varepsilon_{q, p}}$$

- ▶ Use θ as evidence of “tax salience” (is it?)
- ▶ Can we identify markups... ?!?